

AI Compute Accessibility Atlas

Mapping hyperscaler proximity and AI research activity (OpenAlex)

Project Proposal #1 — Revised narrative draft

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Abstract

AI is often discussed as if it were weightless software, but the infrastructure that enables modern AI is highly physical: cloud regions, datacenters, fiber routes, and power systems. Recent evidence shows that frontier-model development has become dramatically more compute-intensive and that governments and firms now treat compute capacity as strategic economic infrastructure rather than background plumbing [8, 9, 11, 13, 12]. This atlas asks whether that infrastructure geography is visible in the geography of AI research.

We measure compute accessibility as the great-circle distance from each city to its nearest AWS, Azure, or Google Cloud region, then join a city-level OpenAlex overlay of AI-related research output and analyze the relationship with two intentionally interpretable spatial models. The main finding is not just that AI cities are closer to compute than other large cities. The gap is large in magnitude, remains negative after controlling for population, and fits a broader story in which cloud infrastructure reinforces already-advantaged urban research corridors. At the same time, the difference between our two model tracks is analytically useful: once we allow for broader regional clustering, the estimated distance penalty becomes smaller, which suggests that compute proximity matters as part of a larger ecosystem of talent, institutions, and connectivity rather than as a standalone driver.

Key terms (plain English)

- **Cloud region:** a named geographic cluster of datacenters operated by a hyperscaler such as AWS, Azure, or Google Cloud.
- **Compute accessibility:** in this project, the straight-line distance from a city to its nearest hyperscaler region. It is a simple proxy for ease of reaching cloud compute, not a direct measure of latency, price, or GPU availability.
- **OpenAlex:** an open scholarly metadata catalog. We use a user-provided OpenAlex-derived city overlay to count AI-related works.
- **Spatial model:** a model that allows nearby cities to resemble one another instead of pretending every city is independent of its neighbors.

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1 Motivation

The geography of AI is usually narrated through firms, labs, and national strategies, but all of those actors depend on infrastructure that is unevenly distributed across space. Public cloud capacity has become a central delivery channel for AI development and deployment, and recent OECD work now treats domestic cloud compute availability as something worth measuring in its own right [10, 11]. The broader backdrop also matters: the Stanford AI Index documents continuing concentration in frontier model development and compute spending, while current energy analysis and industry announcements show how quickly AI infrastructure is scaling [9, 12, 13, 14]. In other words, compute access is not a niche technical issue. It is increasingly part of the economic geography of AI.

That makes a simple descriptive atlas valuable. Before asking causal questions, it is worth establishing three basic facts: where cloud regions sit relative to major cities, whether observed AI research activity is spatially closer to compute than the broader urban system, and how much of that pattern survives once we account for city size and spatial clustering. This is also conceptually important because the cloud is not placeless. Classic work on cloud computing argued that “the cloud” still has real jurisdictional, economic, and geographic locations [16]. Our atlas takes that insight literally and maps it at city scale.

1.1 Research questions

1. How uneven is compute accessibility across the global system of large cities?
2. Are AI research hubs systematically closer to hyperscaler compute than other large cities?
3. After controlling for city size and spatial clustering, does distance to compute still carry an interpretable negative association with AI research output?
4. What does the pattern of close-versus-far cities suggest about the broader geography of AI opportunity?

2 Data

2.1 Cloud regions (AWS, Azure, GCP)

We use a public cloud-region coordinate dataset with provider region codes and approximate latitude/longitude values [2]. These points are not full datacenter footprints, but they are good enough for a first-pass accessibility measure at continental and city-system scale. That choice is substantively defensible because recent OECD work on AI compute availability also treats public cloud capacity as a measurable infrastructure layer rather than an abstract service [10].

Provider	Regions (count)
aws	29
azure	47
gcp	35

Table 1: Hyperscaler cloud regions included (from public region coordinate datasets).

2.2 Cities and population proxy

The comparison frame is the top 8,000 cities in `worldcities.csv`, which provides city names, coordinates, and population values under a CC-BY-4.0 license [3]. We use this frame because the project is trying to compare AI-active cities against the wider urban system, not against a hand-picked list of research hubs.

2.3 AI research overlay (OpenAlex)

The AI activity layer comes from a user-provided OpenAlex-derived overlay of city-level AI-related works [1]. The accompanying topic file records three selected OpenAlex topics (Table 2).

Topic ID	Topic name	Works count (candidate)
T10181	Natural Language Processing Techniques	290539
T12254	Machine Learning in Bioinformatics	276039
T10320	Neural Networks and Applications	249728

Table 2: OpenAlex topic filters used to define ‘AI research’ in this project (from your overlay-generation script).

For the current build, 328 overlay rows are matched onto the `worldcities` frame. Of those, 322 are within the current 75 km match threshold and 6 exceed it but remain in the working sample. That detail matters for interpretation: the model sample is not the full 8,000-city frame, and the overlay should be read as a matched research layer rather than a census of all city-level AI activity.

Important caveat. The repository does not preserve the exact OpenAlex query end date or the exact recency window behind `ai_works_recent`. Cities absent from the overlay should therefore be interpreted as “not observed in this delivered filter” rather than as verified zero-AI locations.

3 Methods

3.1 Main accessibility measure

For each city, we compute the great-circle distance in kilometers to the nearest AWS, Azure, or Google Cloud region. That distance is the core descriptive quantity used throughout the atlas. We also map a simple distance-decay access score for readability, but the interpretation should stay anchored on kilometers because that is the transparent quantity that readers can reason about directly.

3.2 Why the models are simple on purpose

The statistical goal is modest: estimate whether cities that are farther from cloud regions tend to have lower observed AI research output, after accounting for population and broad geographic clustering. We therefore model $\log(1 + \text{AI works})$ using distance measured in 1,000 km units and log population as the only explicit drivers. This is deliberately not a kitchen-sink specification. Keeping the model lean makes the coefficient on distance easier to interpret and keeps the atlas aligned with the project requirement that the analysis remain readable and not over-engineered.

3.3 Two spatial model tracks

We estimate the same basic relationship in two different spatial ways.

Model 1: driver regression plus Gaussian Process residual field. The Gaussian Process (GP) version treats unexplained geography as a smooth surface across the map [5]. Intuitively, this is the “broad regional gradient” model: nearby places are allowed to have similar residual outcomes because they sit in the same larger spatial environment.

Model 2: driver regression plus CAR/GMRF city-neighbor effect. The CAR/GMRF version instead connects each city to nearby cities in a graph and allows neighboring cities to share a latent effect [6, 7]. Intuitively, this is the “local clustering” model: it is more aggressive about absorbing regional and neighbor-based similarity.

Using both is helpful because the difference between them is substantively informative. If distance remains negative in both models, that suggests the result is not just a city-size artifact. If the magnitude shrinks sharply once stronger spatial pooling is introduced, that suggests some of the raw compute gradient is really standing in for wider regional ecosystems. This is an associational exercise, not a causal one. That caution is especially important in urban innovation research, where agglomeration and knowledge spillovers already generate strong geographic clustering [15].

Term	GP+drivers (β)	CAR/GMRF+drivers (β)
Distance to nearest region (per 1000 km)	-0.207	-0.052
City size (log population)	+0.279	+0.309

Table 3: Model coefficient comparison (posterior mean / empirical Bayes). Outcome is $\log(1 + \text{AI works})$.

4 Results

4.1 Compute access is highly uneven across the urban system

Figure 1 maps the city-level distance to the nearest hyperscaler region across the 8,000-city frame.

The first important result is that compute access is not remotely uniform. The median city in the 8,000-city comparison frame sits about 657 km from its nearest cloud region. That number is already telling: for a large share of the world’s major cities, “public cloud” is not locally embedded infrastructure in the way it is in a handful of core corridors. This supports the broader policy concern that AI compute is geographically concentrated rather than naturally ubiquitous [11, 12].

4.2 AI research is much more tightly clustered near compute

Figure 2 shows the OpenAlex overlay, while Figure 3 compares distance-to-compute distributions for the full 8,000-city frame and the matched AI-city sample.

This gap is large enough to matter substantively, not just statistically. The median AI city is about 238 km from its nearest region, compared with 657 km for the broader 8,000-city frame. Put differently, the typical AI city in the overlay is roughly 2.8 times closer to compute than the typical large city in the comparison set. The threshold comparisons tell the same story from another angle:

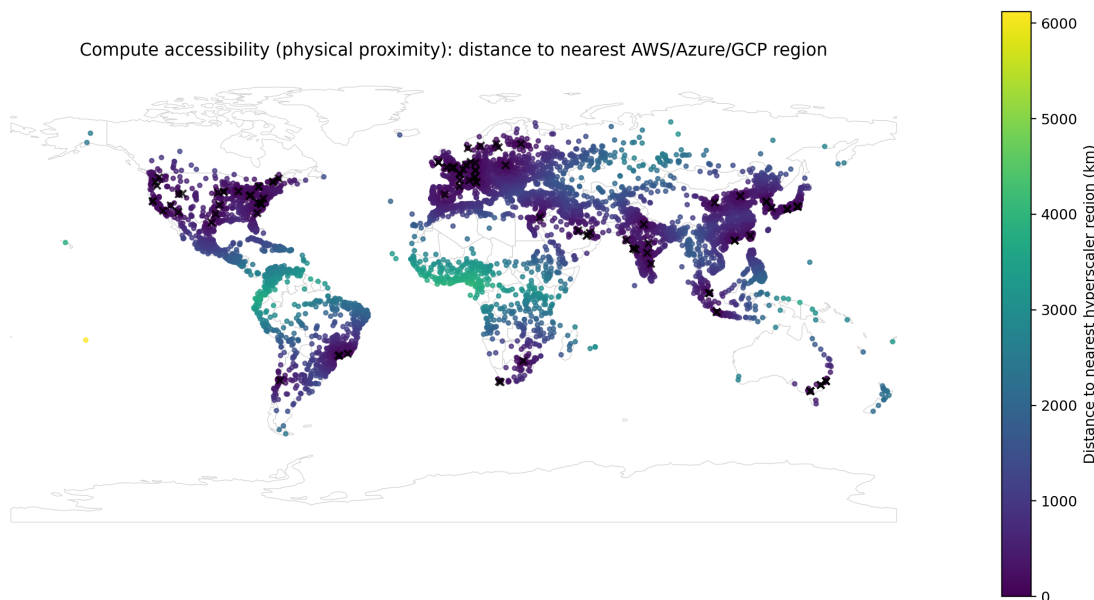


Figure 1: Distance in kilometers from each city to its nearest hyperscaler cloud region. The large gaps outside the main North American, European, and East Asian corridors are the basic spatial fact the rest of the atlas tries to interpret.

72% of AI cities are within 500 km of a cloud region, versus 44% of the full city frame; within 1,000 km, the figures are about 90% and 65%, respectively.

Those percentages change the interpretation. The finding is not merely that a few superstar cities are close to compute. It is that closeness to compute is common across the AI-city sample as a whole. Figure 4 pushes the point further: when cities are weighted by AI works, the mean distance is about 297 km. In other words, the center of gravity of observed AI output is even closer to compute than the unweighted city sample suggests.

My reading of this pattern is that compute proximity behaves less like a luxury amenity and more like a reinforcing production advantage. It likely lowers a bundle of frictions around experimentation, model training, collaboration with cloud vendors, and deployment, even if those mechanisms are not directly measured here. That interpretation is consistent with broader evidence that advanced digital capacity and urban innovation still cluster spatially rather than diffusing frictionlessly [16, 15].

4.3 The mismatch map is useful because it identifies stacked disadvantages

Figure 6 classifies cities by high or low AI activity and high or low compute accessibility using median thresholds.

The quadrant view should not be read as a scorecard of “winners” and “losers.” Its value is that it surfaces cities where multiple disadvantages appear to stack. A city that is both far from compute and low in observed AI output may face several reinforcing constraints at once: cloud access, research density, institutional depth, and international connectivity. Conversely, cities that are high in AI activity despite weaker compute access are analytically interesting because they mark places where other strengths appear to compensate.

Table 4 lists illustrative cases of large cities that are far from hyperscaler regions and have no

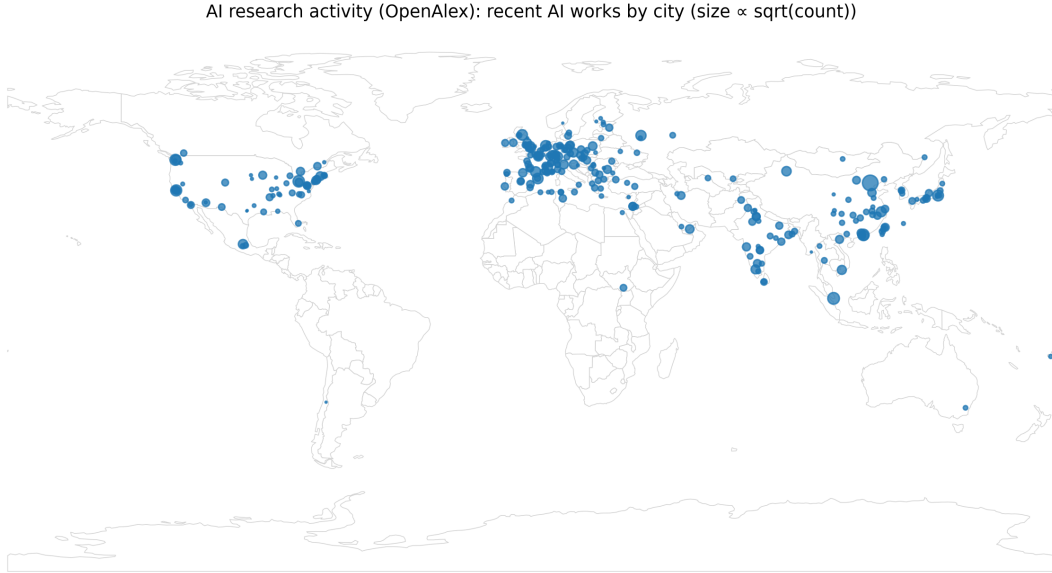


Figure 2: OpenAlex AI works by city. The main visual impression is concentration rather than dispersion: the overlay clusters in the same broad world regions where cloud infrastructure is already dense.

works in the current OpenAlex filter.

4.4 The models say distance matters, but they also show its limits

The two spatial models agree on the sign of the association: greater distance to hyperscaler compute is associated with lower observed AI research output after controlling for population. That directional agreement is important. It suggests that the descriptive pattern is not just an artifact of the fact that big cities are both larger and better served by infrastructure.

The size of the coefficients is where the interpretation becomes more interesting. In the GP model, the distance coefficient is about -0.206 per additional 1,000 km. On a log scale, that corresponds to roughly a 19% lower expected work count for a city that is 1,000 km farther away, holding population fixed. In the CAR/GMRF model, the coefficient is about -0.052, or roughly a 5% lower expected work count per additional 1,000 km.

I do not read that difference as a contradiction. I read it as a bound on the story. The larger GP penalty says there is a substantial broad-scale geography to AI research and compute. The smaller CAR penalty says that once we let nearby cities share more of their fate, some of the apparent compute effect is really regional ecosystem clustering: neighboring cities often share labor pools, institutions, policy environments, and supplier networks. That is exactly why keeping both models is useful. Together they imply that compute accessibility matters, but it is not the only thing that matters, and a narrative that treats compute as the sole driver would overstate the result.

The diagnostics support that cautious reading. The GP model’s 20% holdout RMSE is about 1.54 on the log scale, which is only a rough sanity check rather than a benchmark for deployment. The CAR model fits the observed sample more tightly in-sample, but because its reported summary is more structural than predictive, it should not be treated as a clean forecasting winner. For this atlas, sign stability and interpretability matter more than squeezing out a small prediction gain.

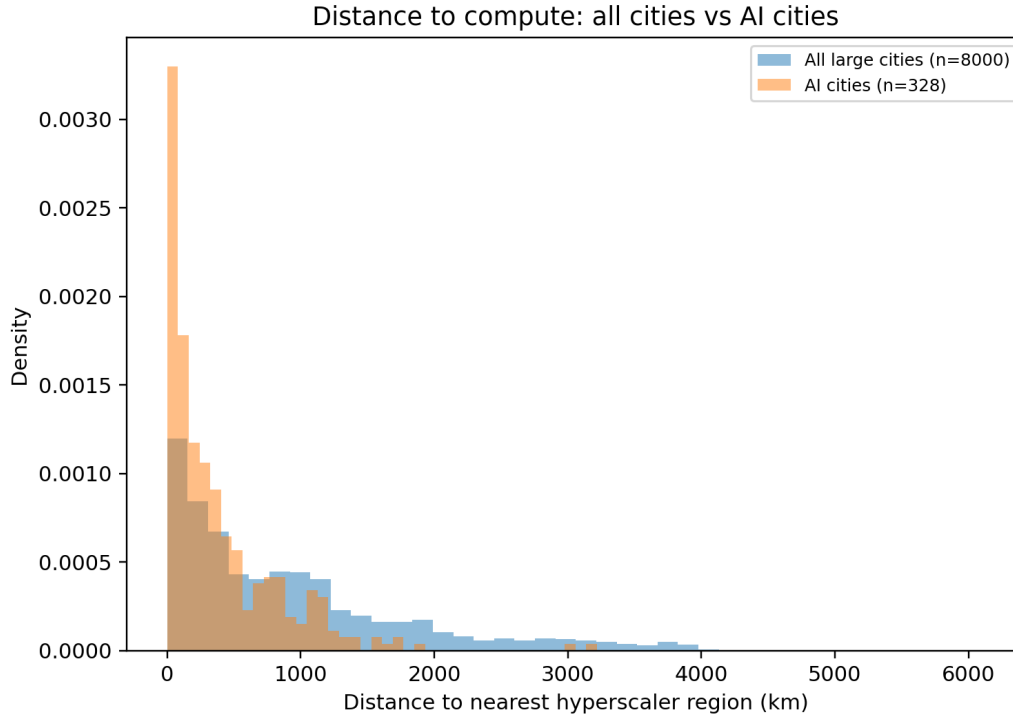


Figure 3: Distance-to-compute distribution for all large cities versus matched AI cities. The left shift is not subtle: the AI-city distribution is concentrated much closer to cloud regions.

4.5 Southeast Asia illustrates the broader logic

Figure 8 zooms into Southeast Asia.

Singapore is a useful anchor case because it shows how compute access can coincide with broader ecosystem advantages rather than replacing them. Its role in the map is not “cloud region therefore AI success.” It is “cloud region plus talent, institutions, and regional connectivity in the same node.” That is the broader interpretation I would carry from the atlas as a whole.

4.6 Top AI cities are rarely far from compute

Table 5 highlights the leading AI cities in the overlay and their nearest cloud regions.

These leading cities are not random points scattered across the globe. Most of them are either region locations themselves or within a few hundred kilometers of one. That does not prove that cloud proximity created those clusters, but it does show that the most visible research centers are usually embedded in places where compute can be reached with relatively low geographic friction.

5 Discussion

The simplest interpretation of the atlas is that AI compute access appears to function as a complementary advantage within already strong urban ecosystems. Cities that combine research institutions, talent, and international connectivity with nearby cloud infrastructure appear repeatedly in the high-output part of the map. Cities that are distant from cloud regions are not doomed to low AI activity, but they are much less likely to show up as major centers in the current overlay.

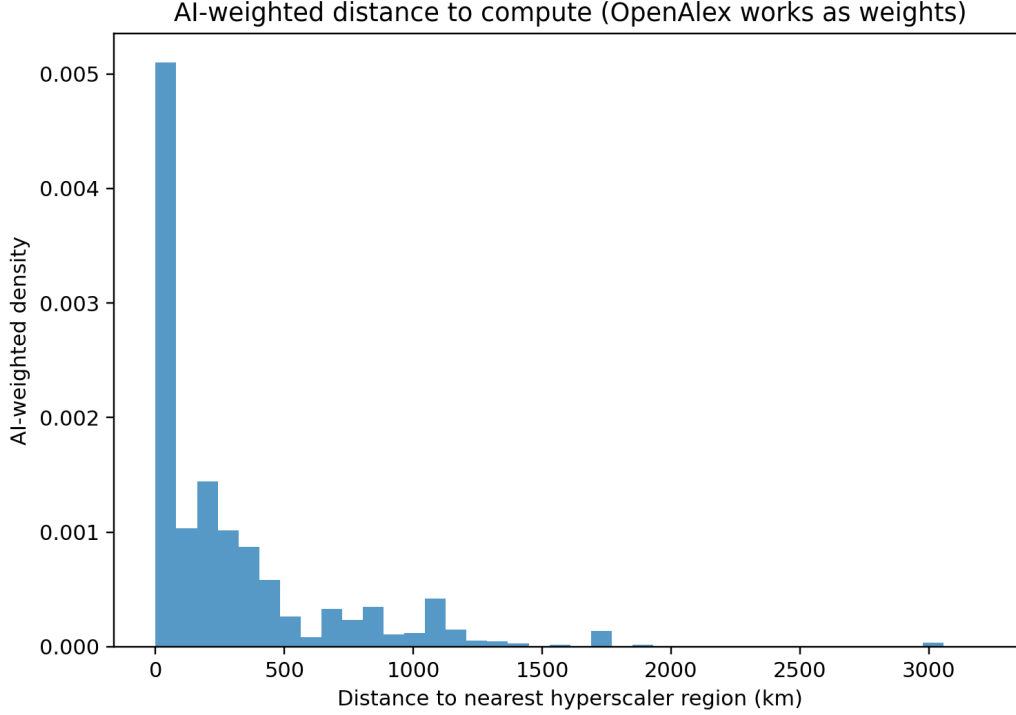


Figure 4: AI-weighted distance distribution. Weighting by research output pulls the center of gravity even closer to cloud regions, which indicates that a disproportionate share of measured AI work is produced in compute-proximate places.

That conclusion matters because it pushes against an overly frictionless story about the cloud. In principle, cloud services can be accessed from anywhere. In practice, the geography of modern AI still looks concentrated and material. Public cloud capacity, power systems, and network connectivity remain unevenly distributed, and those inequalities appear to line up with the research geography seen here [16, 10, 12]. The current infrastructure race around AI datacenters only increases the salience of that point [13, 14].

The most important caution is interpretive discipline. This atlas does not show that opening a cloud region will automatically create an AI hub. The smaller CAR coefficient, the presence of outliers, and the limitations of the OpenAlex overlay all suggest a more modest reading: compute accessibility is one important part of an enabling environment, but it works alongside human capital, institutions, language, funding, and international research networks. That is a meaningful result in its own right because it tells us that compute geography belongs in the conversation, even if it does not settle the conversation.

6 Limitations and extensions

- **Distance is not latency, price, or capacity.** Straight-line proximity is a transparent first-order proxy, but real access depends on routing, service availability, capacity constraints, GPU allocation, regulation, and price. A strong next step would be to add measured network latency or provider-specific availability data.
- **The OpenAlex overlay is informative but not fully documented.** The repository preserves

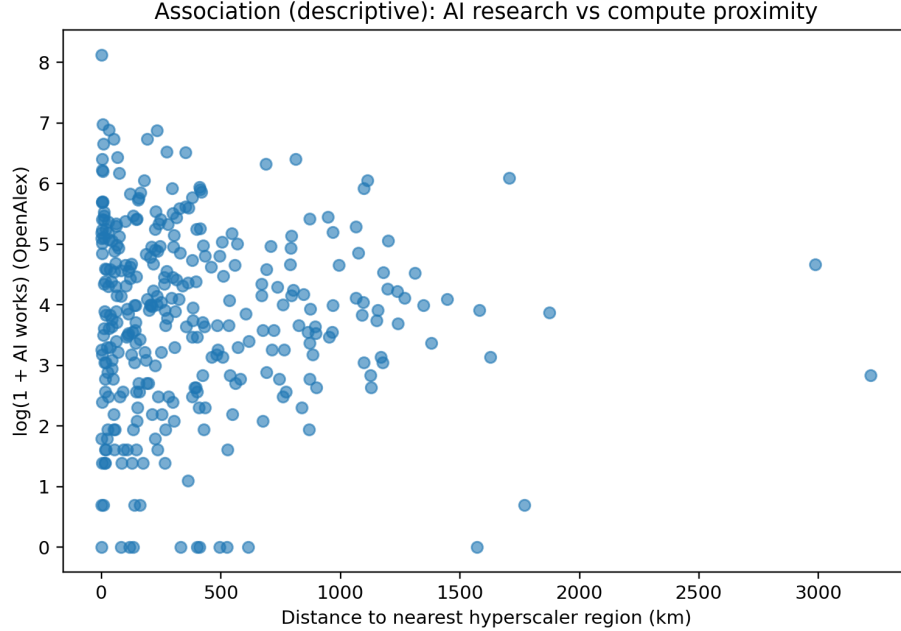


Figure 5: Distance to compute versus $\log(1 + \text{AI works})$. The downward slope is a descriptive pattern only, but it is strong enough to motivate controlled spatial models rather than a purely visual reading.

the delivered files and topic IDs, but not the exact query end date or recency window, so the research layer should be treated as a defensible overlay rather than an exhaustive ground truth.

- **The matched model sample is limited.** The models are fit on 328 matched overlay rows, not the full 8,000-city universe. That is enough to reveal a pattern, but not enough to justify very fine-grained forecasting claims.
- **Spatial models do not create identification.** GP and CAR/GMRF terms help absorb geographic dependence, but they do not turn an observational map into a causal estimate.
- **The surfaces are visualization products.** The 1-degree rasters are useful for communication, but they should not be interpreted as fine-resolution predictive maps.

7 Reproducibility

All code and data-processing steps used for this atlas are included in the repository:

- `src/pipeline.py`: end-to-end data pipeline.
- `outputs/gis/`: GeoPackage, GeoJSON, and GeoTIFF exports used for mapping.
- `outputs/tables/`: model summaries and descriptive outputs.
- `report/main.tex`: the report source file for this atlas narrative.

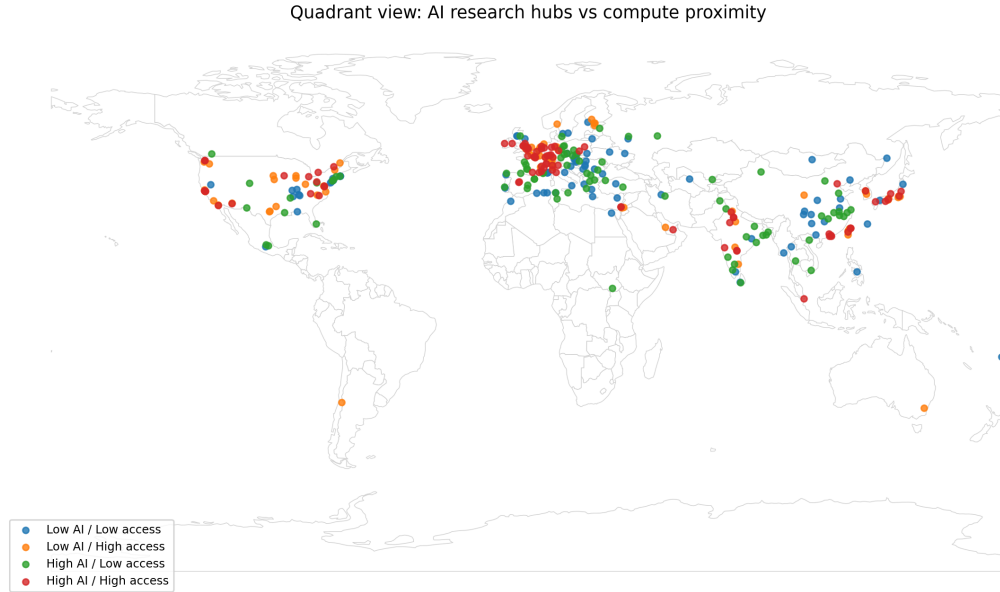


Figure 6: Quadrant view of high or low AI activity versus high or low compute proximity. The most useful reading is diagnostic: it highlights where limited AI presence and weak compute access appear together.

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City	ISO2	Population (approx)	Dist. to nearest region (km)	Nearest provider	Nearest region
Bangkok	TH	17066000	1423	gcp	asia-southeast1
Lagos	NG	15279000	3843	gcp	europa-southwe
Kinshasa	CD	13528000	2750	azure	southafricanort
Lima	PE	9848000	2465	gcp	southamerica-w
Bogotá	CO	9464000	3238	gcp	us-east1
Luanda	AO	8417000	2457	azure	southafricanort
Khartoum	SD	7282000	1787	aws	il-central-1
Dar es Salaam	TZ	6698000	2410	azure	southafricanort
Nairobi	KE	5545000	2870	azure	southafricanort
Abidjan	CI	4980000	3901	gcp	europa-southwe
Santa Cruz	BO	3151676	1821	gcp	southamerica-ea
Addis Ababa	ET	3041002	2275	aws	me-south-1

Table 4: Illustrative ‘AI deserts’: large cities with zero AI works under the selected OpenAlex topic filter, and far from hyperscaler regions (distance $\geq$ 75th percentile).

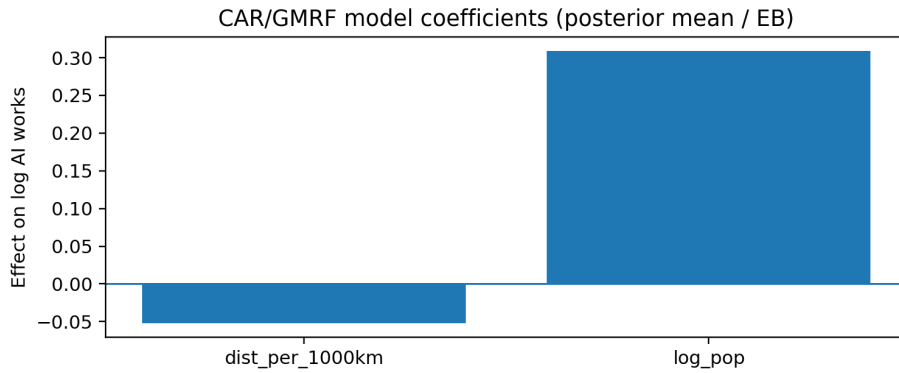


Figure 7: Model comparison for the distance coefficient. Both models keep the sign negative, but the stronger spatial pooling in the CAR/GMRF specification pulls the magnitude closer to zero.

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Southeast Asia focus: AI cities and hyperscaler regions



Figure 8: Southeast Asia focus. Singapore stands out not simply because it is close to compute, but because cloud presence, connectivity, and research capacity appear to reinforce one another in the same place.

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City (OpenAlex)	ISO2	AI works (recent)	Dist. to nearest region (km)	Nearest provider	Nearest region
Beijing	CN	3387	0	aws	cn-north-1
Singapore	SG	1072	4	azure	southeastasia
Shenzhen	CN	978	31	aws	ap-east-1
Pittsburgh	US	965	231	aws	us-east-2
Seattle	US	844	191	azure	westus2
Mountain View	US	841	52	aws	us-west-1
Tokyo	JP	778	6	gcp	asia-northeast1
Barcelona	ES	680	273	azure	francesouth
Edinburgh	GB	673	350	aws	eu-west-1
Toulouse	FR	623	66	azure	francesouth
Moscow	RU	605	811	gcp	europe-north1
Amsterdam	NL	602	1	azure	westeurope

Table 5: Top AI research cities (OpenAlex overlay) and proximity to the nearest hyperscaler region.

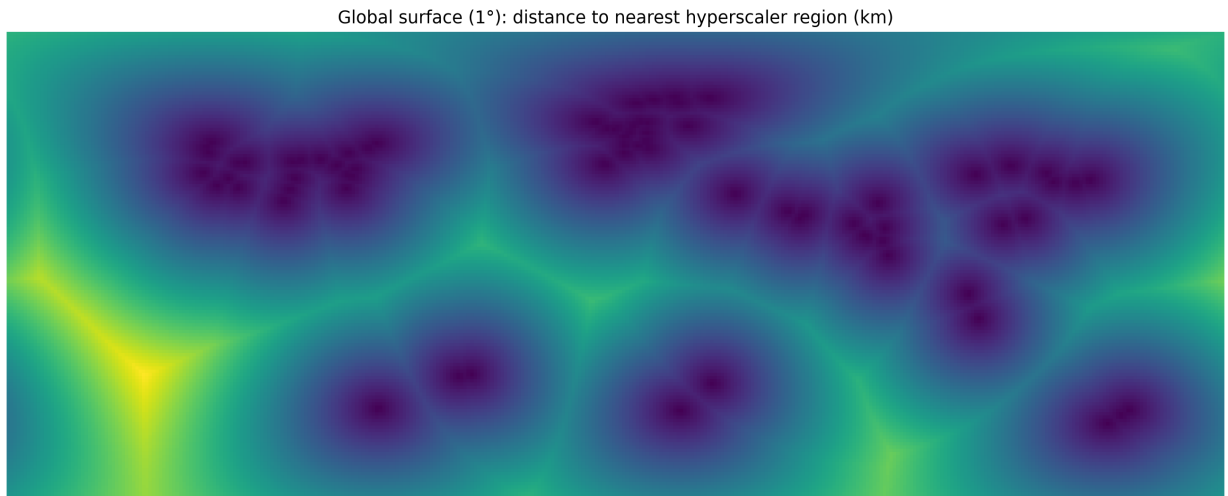


Figure 9: Coarse 1-degree visualization of distance to the nearest hyperscaler region. This is a communication graphic, not a fine-resolution operational map.

Global surface (1°): GP-predicted $\log(1+\text{AI works})$ for a reference 1M-person city

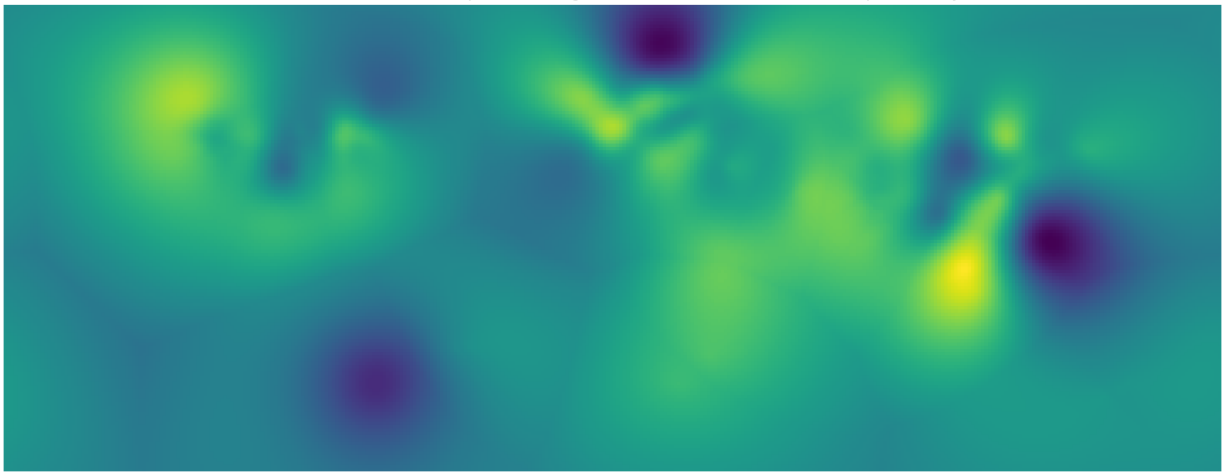


Figure 10: Coarse 1-degree GP visualization for a reference 1M-person city. The purpose is to summarize the broad geography of the fitted relationship, not to make parcel-level or city-block predictions.